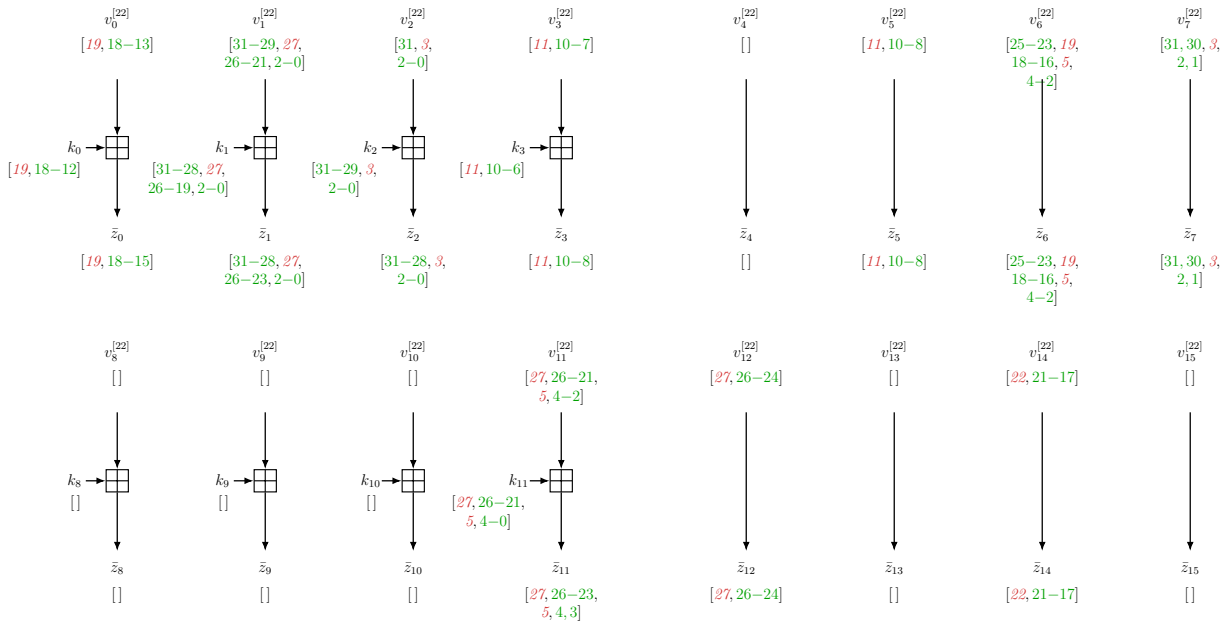




Final key addition for the trail of f_1 in the 5.5-round attack. The trail of 5.5round-f1-bit-propagation.pdf ends with masks on the state $V^{[22]}$. For each key word j that the trail reaches, the figure shows how the mask on the state word $v_j^{[22]}$ (top) is expressed through the keystream word $\bar{z}_j = v_j^{[22]} \boxplus k_j$ (bottom) and the key word k_j (left). The brackets list the bits kept in each mask, in the same colors as in the trail figures. Non-key words are not drawn, because there $\bar{z}_j = v_j^{[22]}$ and the mask carries over unchanged. A **red** key bit is always active, so it only flips the sign of the parity and need not be guessed; the **green** key bits are the ones from which the guessed set K_1 is taken, and the keystream bits are the public inputs of the punctured map g_1 .